IOP Simulation Study − Population 1 (High IOP, omega_r = 0.357)

Figure 1 (Objective iii) − Normal QQ Plot of Standardised IOP Estimates

$z = \dfrac{\hat{\omega}_r - \omega_r}{SE_{JK}}$ | $\omega_r = 0.3571$ | $n_s = 750$ clusters/strat | $k = 4$ HH/cluster | $B = 5000$ reps

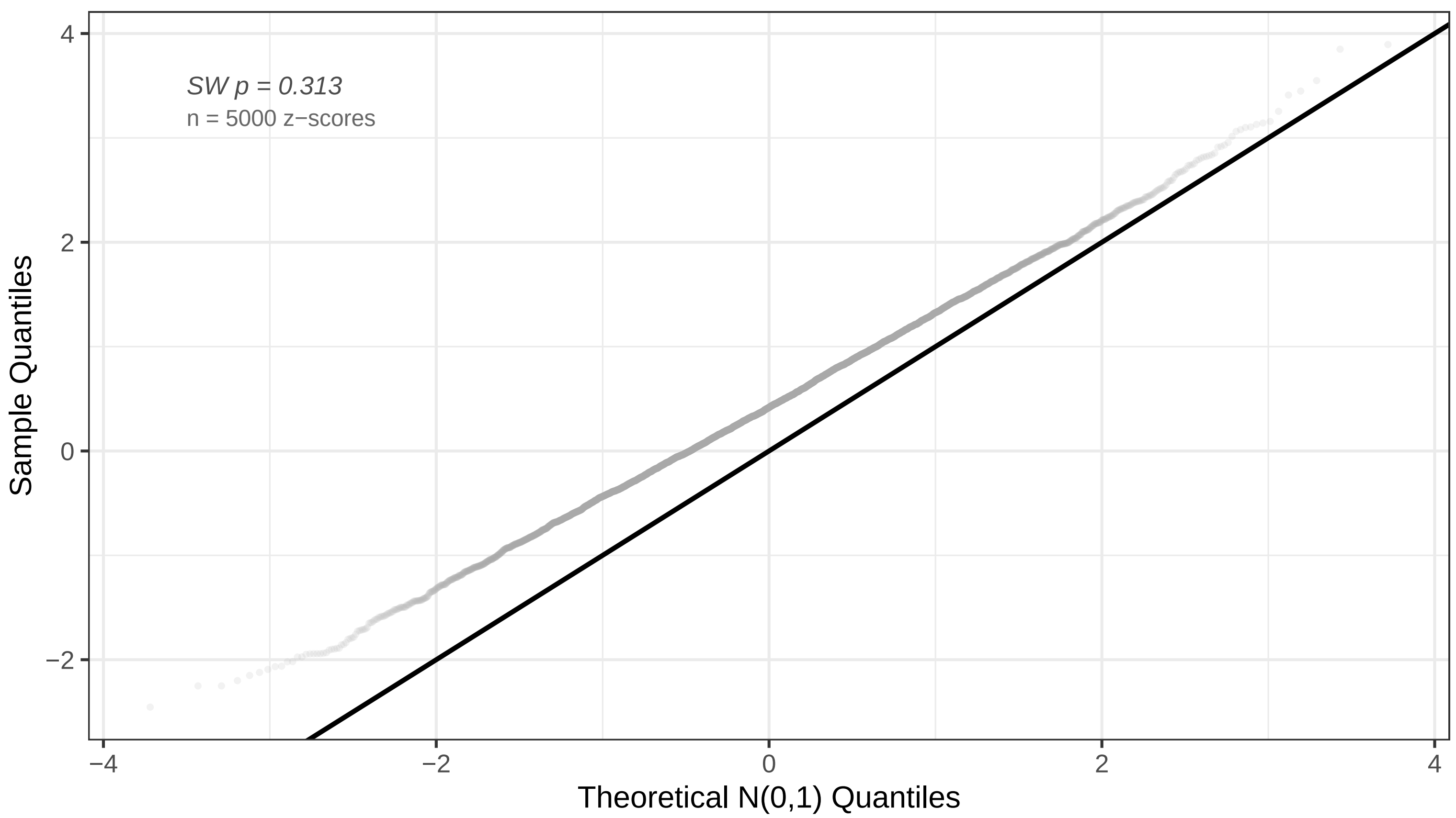

Figure 2 (Objective i) − Sampling Distribution of the Relative IOP Estimator

Bias = 0.00528 | RMSE = 0.01215 | MC SD = 0.01094 | B = 5000 reps

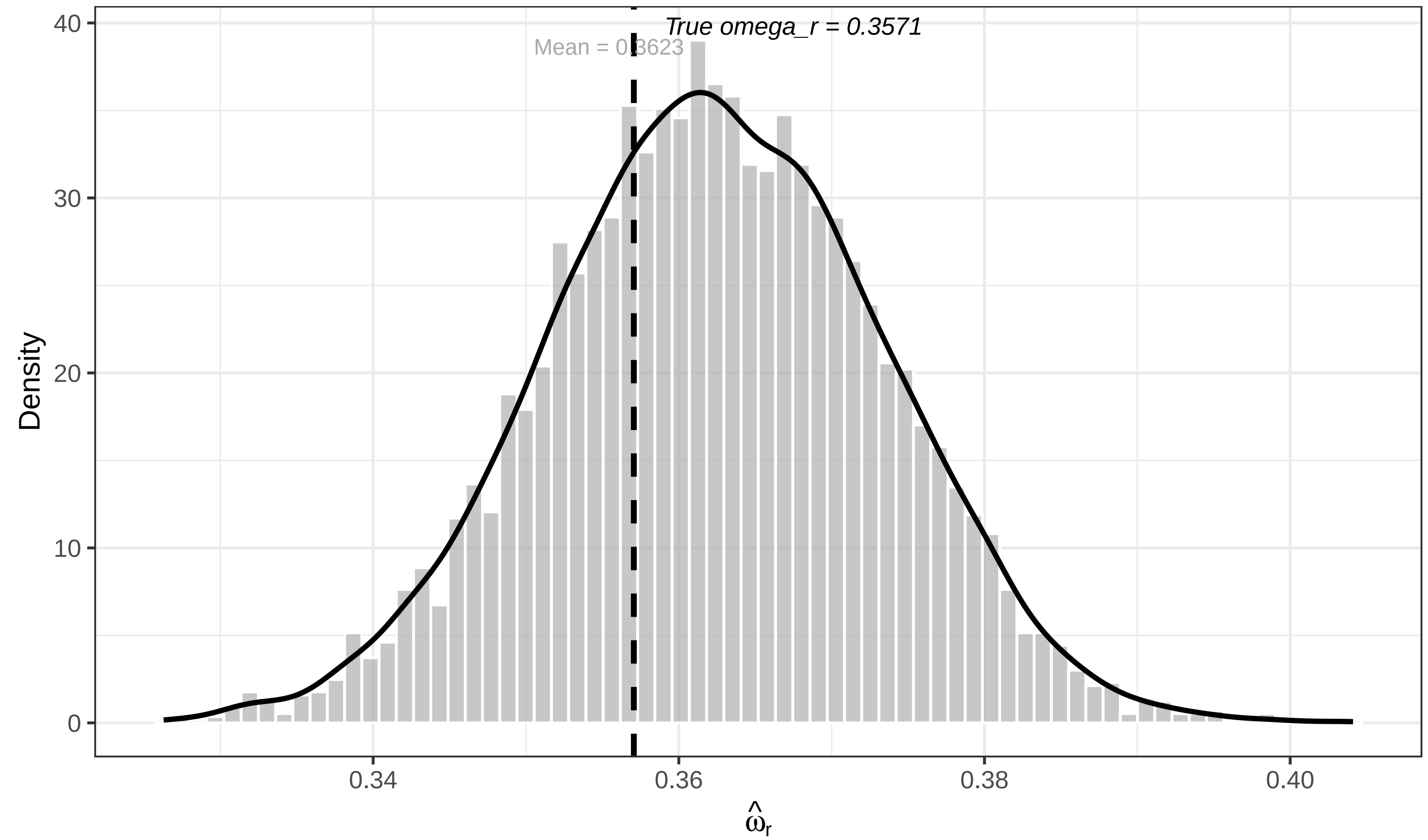

Figure 3 (Objective ii) − Empirical Coverage of 95% Wald Confidence Intervals

$B = 5000$ reps | $n_s = 750$ clusters/strat | $k = 4$ HH/cluster

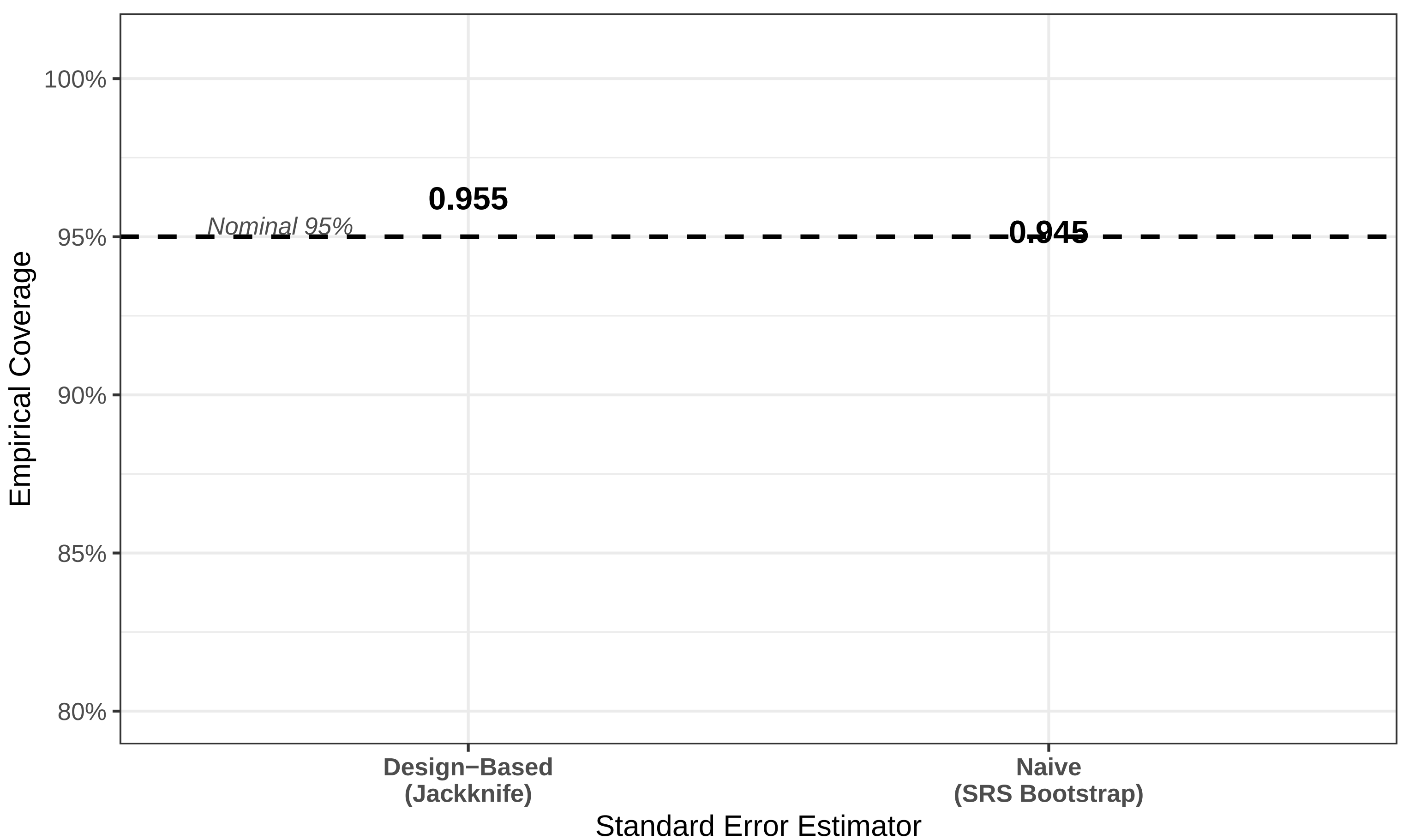

Figure 4 (Objective iv) − Type I Error and Power of the Two−Sample Wald Test

$H_0: \omega_r^{(1)} = \omega_r^{(2)}$ vs $H_1: \omega_r^{(1)} \neq \omega_r^{(2)}$ | $\alpha = 0.05$ | $n_s = 750$ clusters/strat per sample | $B = 2000$ reps